A-A-50796
5 May 82

COMMERCIAL ITEM DESCRIPTION

SIREN, ELECTRONIC

The General Service Administration has authorized the use of this commercial item description.

SCOPE. This commercial item description covers electronic sirens and the general requirements for a commercial electronic siren.

CLASSIFICATION. Siren equipment shall be of the following types, classes, and styles.

Type I. Complete siren. Amplifier with attached microphone, amplifier mounting bracket and hardware, microphone mounting clip, interconnecting wire harness, speaker(s), speaker mounting bracket(s) and hardware.

Type II. Amplifier with attached microphone, amplifier mounting bracket and hardware, microphone mounting clip, interconnecting wire harness.

Class A 12 volts

Style 1 one speaker.
Style 2 two speakers.

Class B 24 volts

Style 1 one speaker.
Style 2 two speakers.

APPLICABLE DOCUMENTS. The following documents form a part of this commercial item description to the extent specified herein. Unless otherwise indicated, the issue in effect on date for bids or request for proposal shall apply.

MIL-W-16878/1 Wire, Electrical, Type B. 105°C, 600 Volts (Insulated, High Temperature).

(Copies of this military specification can be obtained from the procuring activity or as directed by the contracting officer.)

SAE Standard J551 Measurement of Electromagnetic Radiation from Motor Vehicles 20 - 100 MHz.

(Copies of this standard can be obtained from the Society of Automotive Engineers, Inc; 400 Commonwealth Drive, Warrendale, PA 15096.)

ANSI S1.13 - 1971 American National Standard Methods for the Measurement of Sound Pressure Levels.

(Copies of this standard can be obtained from American Standards Institute; 1430 Broadway, New York, NY 10018.)

FSC 6350

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SALIENT CHARACTERISTICS.

1. Design and construction. The maximum overall dimensions of the amplifier, including the mounting bracket, shall not be more than 8 inches wide, 5 inches high, and 9 inches deep. The siren shall be of compact design and sufficiently rugged to withstand shock and stress incidental to use on the vehicle.

2. Modes of operation. The following modes of operation shall be made selectable by control on the front panel of the amplifier assembly. These modes will be clearly marked:

a. PA This mode connects microphone to amplifier for use as public address system.

b. RADIO This mode amplifies calls received on vehicle two-way radio through PA system.

c. MANUAL This mode enables operator to manually control pitch.

d. WAIL Siren will produce continuous automatic cycling of increasing and decreasing sound level at a rate of 7-15 cycles per minute.

e. YELP Siren will produce a continuous automatic cycling of increasing and decreasing sound level at a rate of 100-225 cycles per minute.

3. General requirements. Power-on and/or front panel illumination shall be indicated by a nonglare pilot light. The amplifier assembly with mounting bracket shall be designed for mounting above or under dash. The amplifier mounting bracket shall allow the amplifier to be readily removed for repair purposes. The amplifier's power shall be controlled by an on-off switch located on the volume control, or by a seperate on-off switch. Amplifier output power shall be isolated from the primary input power by a transformer to prevent shorting the battery or damaging the internal components of the electronic siren in case the output becomes grounded. Internal protection shall be provided by a fast-acting fuse (15 amp) or circuit breaker. The electronic siren shall be constructed for use with an electrical ground return and shall not use the chassis for ground return. Siren shall not be required to operate on reversed polarity power connections, but shall not be damaged by such connections. Speakers and mounting brackets when mounted on the outside of vehicle shall be able to withstand a relative wind velocity of 100 mi/hr. The speaker mounting bracket shall be designed to position the speaker on the vehicle centerline (if one speaker applies). Bracket should be designed in such a manner as to prevent easy removal by vandalism. Interconnecting speaker wire shall be 18 AWG wire and a minimum of 15 feet long unless otherwise specified in contract documents. The microphone shall be a magnetic or noise cancelling carbon type. The microphone cord shall be the coil type and shall have an extended length of 8 ft $\pm$ 1 ft.

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4. Operating current requirements.

a. Class A siren.

(1) Maximum standby current shall be .5 amp.

(2) With one speaker (driver) maximum current shall be 6 amps.

(3) With two speakers (drivers) maximum current shall be 12 amps.

b. Class B siren.

(1) Maximum standby current shall be .6 amp.

(2) With one speaker (driver) maximum current shall be 4 amps.

(3) With two speakers (drivers) maximum current shall be 6 amps.

5. PA system. Minimum input required for operation will be .55 volt RMS at 8 ohms.

6. Electromagnetic compatibility. Siren equipment will be compatible with radio communications equipment. Siren equipment will operate properly on vehicles conforming to radiation limits outlined in SAE J551, Measurement of Electromagnetic Radiation from Motor Vehicles.

7. Materials. The amplifier housing shall be made of 18 gauge steel or heavier to provide strength and electromagnetic shielding. Metals used in the manufacture of the siren equipment shall be either corrosion resistant or treated to prevent corrosion. The use of dissimilar metals, especially brass, copper, or steel in intimate contact with aluminum or aluminum alloy shall be avoided. Protective coatings which will crack, chip, peel or scale with age shall be avoided. When materials are used that are nutrients for fungi and they are not hermetically sealed, they shall be treated with a fungicidal agent. The siren shall be wired with a type of wire which conforms to MIL-W-16878/1.

8. Identification markings. Amplifier boxes shall be permanently marked with the manufacturer's name, initial(s), or lettered trademark. The model designation shall be in letters and/or numerals at least .125 inch in height. Drivers shall be marked to denote manufacturer, model, and power rating in letters and/or numerals at least .125 inch in height.

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9 Performance requirements. Siren sound pressure level requirements shall be verified in accordance with methods specified in ANSI S1.13-1971, American National Standard Methods for the Measurement of Sound Pressure Levels.

a. Sound levels. Measurements to be made with 11 ohms, 75 watts speaker(s)/driver(s).

b. PA sound output. Minimum 112 db at 10 feet with one speaker. Minimum 116 db at 10 feet with two speakers.

c. Siren sound output. Minimum 114 db at 10 feet with one speaker. Mimimum 118 db at 10 feet with two speakers. Siren output shall be a symmetrical square wave output. The lowest fundamental frequency shall be between 250 Hz and 750 Hz. The frequency must rise at least 900 Hz.

d. Wail. Siren will produce continuous automatic cycling of increasing and decreasing sound level at a rate of 7-15 cycles per minute.

e. Yelp. Siren will produce continuous automatic cycling of increasing and decreasing sound level at a rate of 100-225 cycles per minute.

f. Audio frequency response. It shall be flat within 3 db from 250 to 4000 Hz at rated power. 1000 Hz shall be used as the zero db reference.

10. Environmental requirements. The siren shall be operable in the temperature range -40°C (-40°F) to 70°C (158°F) and shall be designed to withstand relative humidities up to 100 percent. The siren shall be designed to withstand vibration for a period of at least 1 hour at 12 Hz $\pm$ 1 Hz, .3 cm (.125 in) peak to peak displacement. All speakers shall be weather proof.

11. Workmanship. The finished siren or individual accessories shall be free from burrs, slivers, sharp edges, dents, and fractures. Bolts, nuts, and screws shall not be missing, loose, fractured, or stripped.

CERTIFICATION. The contractor shall certify that the product offered conforms to the requirements specified herein and to the producers' own drawings, specifications, standards, and quality assurance practices, and is the same as the product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first article delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Reclaimed Materials. The manufacturer shall utilize recovered materials to the maximum extent practicable.

Packaging & Packing. The packaging and packing shall be in accordance with normal commercial practice and shall assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with the National Motor Freight Classification and Uniform Freight Classification. Marking shall be as specified in the contract or order.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 99

Review activity:
Army - ME

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